The Journal of Systems and Software 237 (2026) 112836

Contents lists available at ScienceDirect

The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss

A systematic literature review on model-based requirements engineering ⋆

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ARTICLE INFO

Editor: Dr. P. Lago

Keywords:
Model-based requirements engineering
Requirements modeling
Software engineering

ABSTRACT

Over the past decade and a half, significant methodological and technological advancements have been made in Model-Based Requirements Engineering (MBRE). This study conducts a survey to investigate the evolutionary trends in MBRE research from 2010 to 2025, with a focus on five critical dimensions: modeling methodologies, requirements engineering activities, quality assurance mechanisms, application domains, and core research themes. Through rigorous selection criteria, 109 high-quality papers were identified from major academic databases including IEEE Xplore and ScienceDirect. Statistical analysis and systematic categorization were employed to synthesize the extracted data. The primary contribution of this paper lies in providing a comprehensive synthesis of recent MBRE developments, while also identifying theoretical limitations and practical challenges in current research. These findings offer valuable insights for researchers to pinpoint critical knowledge gaps and assist practitioners in selecting appropriate methodologies for specific engineering contexts.

1. Introduction

Model-Based Requirements Engineering (MBRE) has become an important methodology in the fields of software and systems engineering, emphasizing the use of models to capture, analyze, and manage requirements. With the increasing complexity of systems and the demand for more efficient, scalable, and reliable engineering practices, MBRE has developed significantly over the past decade and a half. Although numerous studies on MBRE have summarized its key achievements, provided insights into the field, and outlined challenges in various directions, to the best of our knowledge, most of these studies focus on MBRE within specific domains. For example, Rashid et al. conducted a systematic literature review on model-based requirements trends in embedded design (Rashid et al., 2016). Li et al. presented the challenges of context-aware requirements modeling through a systematic literature review (Li et al., 2018). Yang et al. conducted a systematic literature review on requirements modeling and analysis for adaptive systems (Yang et al., 2014). Whittle et al. conducted two rounds of semi-structured interviews with 39 industrial MDE practitioners, analyzed data to construct and validate a taxonomy of MDE tool-related issues (Whittle et al., 2017). Ameller et al. conducted an interview survey with MDD practitioners from 18 companies in 6 European countries, revealing the current status of NFRs management in MDD (Ameller et al., 2021). There has been no systematic analysis of the research trends across the entire MBRE field or a categorization and evaluation of the corresponding requirements engineering activities and application domains. As a result, there is still no clear understanding of which research areas have been focused on, where the results have been published, the extent to which each modeling method or requirements engineering activity has been studied, how the methods have been evaluated, the quality of the studies, and the most active research topics.

The objective of this paper is to systematically investigate the research literature in the field of MBRE, summarize the current research trends, categorize the modeling methods and relevant requirements engineering activities, analyze research quality and application domains, assess the quality of the current studies, and identify the most active research topics.

The rest of the paper is structured as follows: Section 2 briefly introduces the survey methodology and the research questions, adopting the 5W1H method in Zhou et al. (2020); Section 3 reports the results of the literature selection and its categorization; Section 4 provides a comprehensive classification and analysis of existing MBRE research; Section 5 presents the modeling techniques used in the selected studies and evaluates and compares them; Section 6 assesses the quality of the selected studies; finally, Section 7 concludes the paper and discusses its limitations and future work.

⋆ Editor's note: Open Science material was validated by the Journal of Systems and Software Open Science Board.

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https://doi.org/10.1016/j.jss.2026.112836
Received 3 June 2025; Received in revised form 9 January 2026; Accepted 2 March 2026
Available online 7 March 2026

2. Survey method

This section outlines our literature review methodology by defining our research questions, search databases, keywords, and search process. It will also provide a detailed description of the literature selection steps to ensure the systematic and effective nature of the entire process.

2.1. Research questions

The research questions that we intend to answer in this study are defined as follows.

- RQ1(Why):What are the primary objectives and driving motivations behind adopting model-based requirements engineering?
- RQ2(When):What does the historical development trajectory of the study of MBRE look like?
- RQ3(Where):Which countries/institutions lead MBRE research? Which publication venues are primary channels for MBRE studies?
- RQ4(Who):Which institutions and researchers have made outstanding contributions in the field of MBRE?
- RQ5(What):What are the predominant technical approaches in MBRE, and what are their comparative strengths and weaknesses?
- RQ6(How):How are the qualities of the selected studies?

2.2. Research method

The following academic databases have been selected to perform the survey on model-based requirements engineering. They were selected due to their established prominence and scholarly influence within the domains of information technology and software engineering.

- IEEE Xplore (https://ieeexplore.ieee.org/Xplore/)
- Springer Link (https://link.springer.com/)
- Science Direct (https://www.sciencedirect.com/)
- Google Scholar (https://scholar.google.com/)

The following search keywords are used to find relevant studies in papers' title, abstract and index terms. "model-based requirements engineering" OR "requirements modeling" OR "model-driven requirements engineering".

Upon running the queries on the selected databases, we first performed an initial screening based on titles to exclude literature that was obviously irrelevant to the research topic (e.g., studies from unrelated disciplines). The remaining records were then imported into the Zotero reference management software, which automatically identified and removed duplicate entries. This preliminary phase resulted in a refined dataset of 437 unique papers for subsequent formal evaluation.

To ensure the review aligns with modern MBRE practice and captures the field's mature development phase, we focused on the period from 2010 to 2025. This timeframe was chosen for two key reasons: first, it prioritizes timeliness, as post-2010 research reflects widespread industrial adoption and addresses current practitioner needs; second, it corresponds to critical MBRE technical advancements, such as the maturation of SysML. To ensure strict methodological traceability, this time constraint was formally applied as an Exclusion Criterion (EC6) during the screening process.

To ensure the review aligns with modern MBRE practice and captures the field's mature development phase, we restricted the literature publication time to 2010–2025 (with the search conducted in January 2025). This timeframe was chosen for two key reasons: first, it prioritizes timeliness–pre-2010 studies often focus on early theoretical exploration of MBRE, while post-2010 research reflects widespread industrial adoption (e.g., compliance with contemporary safety standards) and addresses current practitioner needs; second, it corresponds to critical MBRE technical advancements, such as the maturation of SysML (e.g., SysML v1.3 in 2012, v1.6 in 2019) which standardized modeling for complex systems and became a cornerstone of modern MBRE. During this query, literature obviously irrelevant to the research topic was filtered out via title screening, resulting in 437 papers for subsequent evaluation. This step excluded topic-deviant works at the earliest stage to narrow the scope efficiently.

We use the following inclusion criteria for selecting the research studies by personally reading. Research articles that meet all these criteria will be selected for the survey.

1) INCLUSION CRITERIA

- IC1) Studies with a clear focus on some aspects of MBRE.
- IC2) Studies providing solutions for needs and problems of MBRE.
- IC3) Studies subject to peer review (e.g. journal article, conference paper, workshop proceeding, and book chapters).
- IC4) Studies written in English.
- IC5) Studies available as full text.

We adopt the following exclusion criteria to screen out undesirable research studies, and any research article meeting one of these criteria will be filtered out. As one of these criteria, we excluded secondary systematic literature reviews (SLRs) from the final dataset–this is to avoid data redundancy with primary studies and ensure methodological consistency in trend analysis. That said, high-quality prior SLRs were consulted during the study design phase to inform our search strategy, inclusion criteria, and analytical framework.

2) EXCLUSION CRITERIA

- EC1) Studies not focusing on requirements modeling or solving some problems in requirements engineering with models.
- EC2) Studies that do not provide any conceptual, formal, or technical solution for the proposed approach to MBRE.
- EC3) Secondary or concluding studies (e.g. systematic literature reviews, surveys).
- EC4) Studies with unclear contributions.
- EC5) Studies in the form of white papers, tutorial papers, short papers, poster papers, or manuals.
- EC6) Studies published outside the selected timeframe (2010–2025).

After database queries, the selection process was conducted in two main stages (as illustrated in Fig. 1):

Stage 1: Criteria-Based Screening. The 437 papers were assessed one by one against the predefined inclusion criteria (IC1–IC5) and exclusion criteria (EC1–EC6). This rigorous screening involved reviewing titles, abstracts, and full texts where necessary. Papers were excluded if they fell outside the time window (EC6), lacked relevance to the MBRE domain (EC1), or failed to meet quality standards (EC2, EC4). After this evaluation, 192 candidate papers that met both topic relevance and academic quality requirements were retained.

Stage 2: Categorization and Re-selection. To ensure the selected studies strongly represented the core research directions, we preliminarily classified the 192 candidate papers into four content-based categories. Within each category, we conducted a deeper content assessment, prioritizing studies with higher relevance, clear methodological rigor, and significant contributions to the identified sub-domains. Papers that were relevant but deemed marginal or superseded by more comprehensive studies within the same category were filtered out. This final refinement resulted in the identification of 109 target papers for this survey.

- Type A (30 papers): Theoretical research on MBRE concepts or foundational principles.
- Type B (58 papers): Methodological research (e.g., novel modeling approaches, tool development for MBRE).
- Type C (11 papers): Application practice of MBRE in specific domains (e.g., aerospace, healthcare systems).
- Type D (10 papers): Empirical research (e.g., case studies, experimental validation of MBRE methods).

Fig. 1 shows our search and screening process. It is noteworthy that the screening process was completed independently by a single author

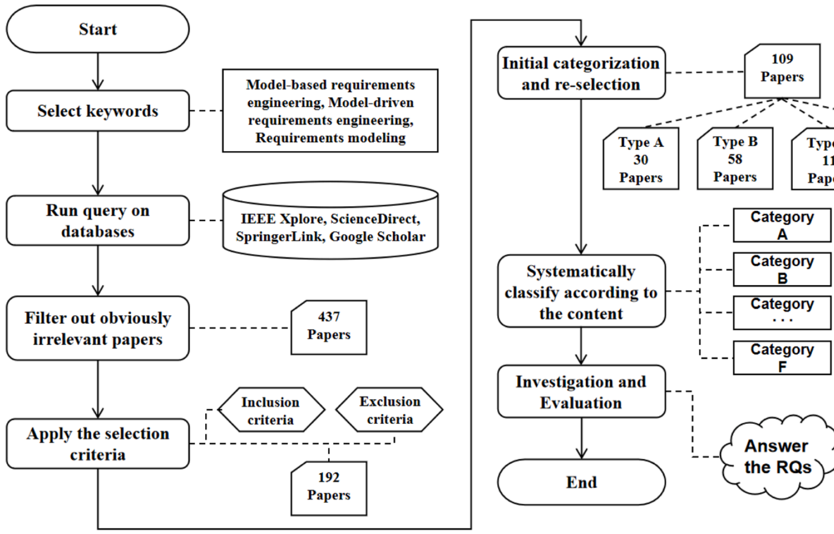


Fig. 1. Overview of the review process.

in January 2025. Thus, inter-rater bias is not a concern here, but literature search results are subject to temporal constraints. After identifying the papers, we analyzed them in detail and recorded bibliographic information (e.g., author affiliation, publication year, publication type, publication region, and publisher), specific content (e.g., research topics, modeling languages, fields of application, and number of citations) for each selected papers.

To ensure the reproducibility of this study, the paper screening and data extraction sheets used for literature collection and organization, along with all scripts employed for figure generation, have been deposited in a public GitHub repository. Access the repository at: https://github.com/Anonymous-161/MBRE.

3. Results

In this section, our objective is to address the following research question: RQ1. What are the primary objectives and driving motivations behind adopting model-based requirements engineering? To answer this question, we initially categorize the selected studies and briefly describe a few studies from each category.

1. REQUIREMENTS MODELING APPROACH
   Development and application of methodologies, modeling languages (e.g., UML/SysML) and tools to support systematic requirements expression.
   (a) Requirements modeling approach
   Ali et al. proposed a goal-oriented RE modeling and reasoning framework for systems operating in varying contexts. Eridaputra et al. proposed a generic requirement model for Big Data application which offers the use of i× Framework and Knowledge Acquisition automated System (KAOS) (Ali et al., 2010). Yang et al. proposed a requirements model for ML-enabled software systems and a modeling process for this model based on an extension of UML (Yang et al., 2024).
   (b) Requirements modeling approach (Language)
   Alférez et al. presented a multi-view composition language for SPL requirements, the Variability Modeling Language for Requirements (VML4RE) (Alférez et al., 2010). Mouratidis et al. presented a novel security modelling language and a set of original analysis techniques, for capturing and analysing security requirements for cloud computing environments (Mouratidis et al., 2020). Wang et al. developed a requirement modeling language called SPARDL for modeling and analyzing periodic control systems (Wang et al., 2010).
   (c) Requirements modeling approach (Tools)
   S.Supakkul and Chung developed a multi-notational requirements modeling toolkit (RE-Tools), supporting many leading requirements modeling notations (Supakkul and Chung, 2012). Yang et al. developed RM2PT: a tool for generating prototypes from requirements models automatically (Yang et al., 2019). Bao et al. proposed a CASE tool named RM2Doc, which can automatically generate ISO/IEC/IEEE 29148-2018 conformed requirements documents from UML models without any templates (Bao et al., 2022).
2. REQUIREMENTS TRACEABILITY AND MANAGEMENT
   Establish the correlation between requirements and design, testing and other phases to ensure change traceability and consistency maintenance.
   Goknil et al. provided the classification of requirements changes based on structure of a textual requirement with formal semantics (Goknil et al., 2014). D.Pandey et al. proposed an effective requirements engineering process model to produce quality requirements for software development (Pandey et al., 2010). Holder et al. presented a method in order to combine the advantages of a conscious requirements management process and graph-based design languages (Holder et al., 2017).
3. REQUIREMENTS ANALYSIS
   Resolve logical conflicts between requirements and resource constraints through conflict detection and prioritization.
   Brace and Ekman presented the checklist-oriented requirements analysis modelling (CORAMOD) approach (Brace and Ekman, 2014). Liao et al. used Kano requirement model to find essential demands, clarify customers' demands, and utilizes extension innovation methods to establish models for the products, which have different characteristics (Liao et al., 2015). Ma et al. proposed a multicriteria requirement ranking method based on uncertain knowledge representation and reasoning (KRR) (Ma et al., 2024).
4. REQUIREMENTS VERIFICATION AND VALIDATION
   Utilize formal methods (e.g., model testing) with user testing to validate requirements correctness and compliance.
   Aceituna et al. proposed a model-based requirements verification method called NLtoSTD, which transforms NL requirements into a state transition diagram (STD) that can be verified through automated reasoning (Aceituna et al., 2011). Granda et al. presented

CoSTest, a tool that supports the validation of Conceptual Schemas by using testing (Granda et al., 2017). Bacquet et al. presented a system requirements conceptual model with requirements writing patterns that enable the systems requirements modeling and V&V, along with a comparative analysis with existing concepts in the literature (Bacquet et al., 2024).

5. REQUIREMENTS ENGINEERING AND FULL LIFECYCLE INTEGRATION

   Embedding the requirements model into the entire development process (e.g. Agile development) to automate requirements-driven iterations.

   Wanderley et al. presented the SnapMind framework to make the requirements modelling process more user-centered, through the definition of a visual requirements language, based on mind maps, model-driven and domain specific language techniques (Wanderley et al., 2014). Windisch et al. presented an approach providing activity-based views in conjunction with activity steps based on a consistent ontology for the description of product requirements and verification activities (Windisch et al., 2022).

6. EVALUATION AND VALIDATION

   Evaluate the practical efficacy and limitations of requirements engineering methodologies through industrial cases and quantitative metrics.

   Abrahão et al. presented a method for evaluating the quality of requirements modeling methods based on user perceptions (Abrahão et al., 2011). Hadar et al. compared the comprehensibility of requirements models expressed in two visual modeling languages: Use Case, which is scenario-based, and Tropos, which exploits goal-oriented modeling (Hadar et al., 2013). Kannan et al. bridged the gap between current Automotive MES and industry standards by following a model-based requirements engineering approach along with a gap analysis process (Manoj Kannan et al., 2017).

After the above careful categorical analysis of selected studies, we summarize the following points to answer RQ1.

Core Objectives

MBRE aims to transform ambiguous natural language requirements into computable, traceable, and verifiable model entities through structured modeling systems (e.g., formal modeling languages, domain-specific frameworks) and model-driven automated toolchains. Its core objectives include:

1. Requirement Formalization and Structuring:

   Use unified modeling languages (e.g., extended UML/SysML) or domain-specific modeling frameworks to convert unstructured requirements into visual, hierarchical models, eliminating textual ambiguity and establishing a shared semantic foundation across interdisciplinary stakeholders.
2. Full Lifecycle Traceability Management:

   Leverage bidirectional traceability tools (e.g., model-requirement linking engines) to create dynamic mappings between requirements and downstream phases (design, testing), enabling automated change impact analysis and consistency maintenance to avoid cross-phase information gaps.
3. Requirement Validation and Optimization:

   Employ formal analysis tools (e.g., conflict detection algorithms, priority ranking models) to identify logical contradictions or resource constraints in requirements, and validate their reasonableness through user feedback and scenario simulation (e.g., prototype generators) to mitigate early-stage development risks.
4. Development Process Integration and Automation:

   Embed requirement models into agile/waterfall workflows via model-driven development platforms to automate end-to-end processes–from modeling to code generation and test case derivation–thereby accelerating iteration cycles and engineering efficiency.

Driving Motivations

MBRE's adoption is driven by three core challenges in modern systems engineering: complexity, compliance, and efficiency bottlenecks:

1. Coping with Surging System Complexity:

   Dynamic, distributed requirement scenarios in emerging technologies (AI, IoT, cloud computing) exceed the capacity of traditional text-based descriptions to capture multi-dimensional (functional/performance/security) requirement interdependencies. Visual modeling simplifies cognitive load, managing hierarchical structures and dependencies.
2. Meeting Compliance and Audit Requirements:

   Safety-critical domains (aerospace, medical devices) impose strict regulatory demands (e.g., ISO 26262, IEC 61508) for requirement traceability and verifiability. MBRE ensures compliance and evidence chain generation through formal semantic models and automated audit tools.
3. Overcoming Traditional Process Inefficiencies:

   Manual requirement management suffers from high error rates (omissions, misinterpretations) and low reusability. MBRE reduces human intervention via executable model features (e.g., automated test case/prototype generation) and cross-phase automation, supporting rapid iteration and requirement evolution.
4. Advancing Engineering Methodology Scientific Rigor:

   Traditional requirements engineering relies on experience, lacking quantitative analysis and empirical validation. MBRE transforms requirements engineering from an "art" to a measurable, reproducible science via data-driven modeling methods (e.g., knowledge-based requirement ranking, statistically validated model evaluation).

4. Publication trends

In this section, we will introduce the publication trends of the research in MBRE, including publication year, region, venue, and research communities to answer RQ2-RQ4. Besides, we also conducted a statistical analysis of the research topics of selected studies and analyze MBRE research trends in various domains through a 2D heat map of Type-Application Domain

4.1. Publication year

In this section, we aim to answer the following research questions: RQ2. What are the key developmental phases and evolutionary characteristics in MBRE research? (This research question will also be discussed further in Section 4.4 Research Topics and Trends.)

Fig. 2 shows the publication year of selected papers, revealing volatility in MBRE literature over the last 15 years, particularly between 2010 and 2019, with fluctuations in the number of releases each year. These variations may reflect cyclical shifts in research focus or academic discussions. From 2020 to 2023, a significant decline in publications occurred, likely due to the impact of the COVID-19 pandemic, which could have delayed research or altered resource allocation. However, the rebound in 2024 suggests a recovery and potential emergence of new research directions. It is important to note that the search was completed in 2025, so studies published that year are only partially represented.

Additionally, it is interesting that the figure shows a yearly alternation pattern in the number of published literatures. To clarify the cause of this fluctuation, we conducted targeted verifications: (1) Regarding conference/workshop factors: We reviewed the hosting regions and holding cycles of major field-related conferences during the study period, and found no correlation between high/low literature output years and conference locations or biennial cycles. (2) Regarding researchers' work cycles: We analyzed core authors' publication timelines and research project cycles, and similarly did not observe a consistent overlap between output fluctuations and academic years or project conclusion stages.At present, the specific driver of this fluctuation remains

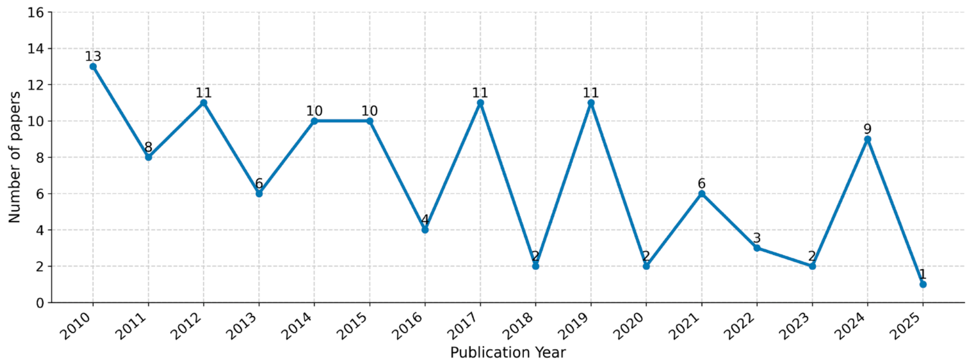


Fig. 2. Publication year of selected studies.

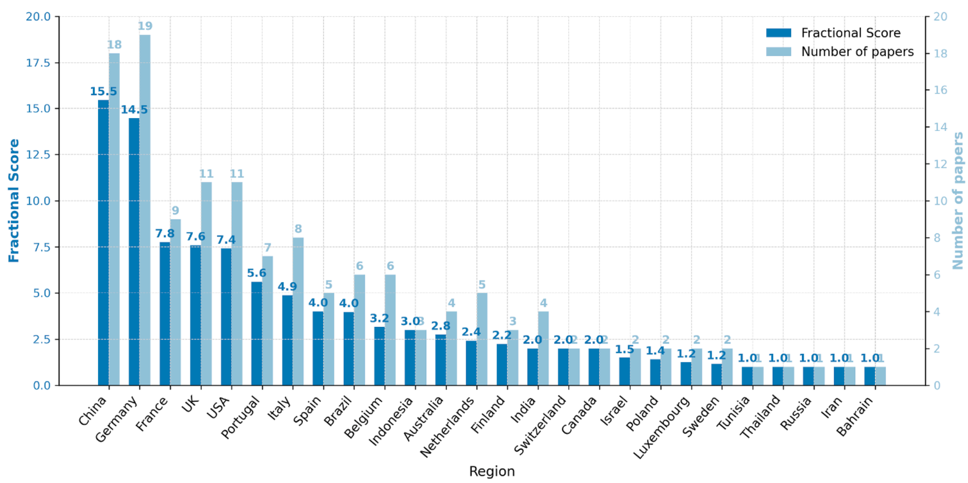


Fig. 3. Top active countries in MBRE research ranked by fractional score (scores ≥ 1.0). The chart displays both the Fractional Score (left axis/dark bar) representing weighted contribution and the Number of Papers (right axis/light bar) representing total participation.

unclear, which may be attributed to the superposition of multiple complex factors (e.g., periodic technical hotspots, database collection efficiency differences, or industry demand fluctuations) that are difficult to isolate with the current data. Future research can expand the analysis scope (e.g., adding more databases or industry demand data) to further explore this phenomenon.

4.2. Publication region and venue

In this section, we aim to answer the following research questions: RQ3. Which countries/institutions lead MBRE research? Which publication venues are primary channels for MBRE studies?

To comprehensively assess the contributions of different regions and institutions, we employed a fractional counting method alongside the total paper count. This approach assigns a weight of 1/N to each author of a paper with N authors, ensuring that credit is fairly distributed among all collaborators rather than attributed solely to the first author. Consequently, we present two indicators for each country in Fig. 3: the Fractional Score (representing the weighted contribution magnitude) and the Number of Papers (representing the breadth of participation).

As shown in Fig. 3, China and Germany emerge as the two dominant leaders in the MBRE domain. China ranks first in terms of weighted contribution with a fractional score of 15.5 (participating in 18 papers), indicating a strong output of lead-authored research. Germany, while ranking second in fractional score (14.5), holds the highest number of total participating papers (19), suggesting a very high level of activity and extensive collaboration within the community.

Following the top tier, France (Score: 7.8, Papers: 9), the UK (Score: 7.6, Papers: 11), and the USA (Score: 7.4, Papers: 11) form the second tier of contributors. It is noteworthy that while the UK and USA participated in more papers than France, France achieved a slightly higher fractional score, implying a higher concentration of primary contributions relative to its total volume.

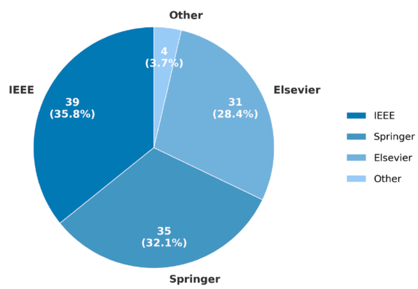
Other active countries include Portugal (5.6), Italy (4.9), and Spain (4.0). From a continental perspective, Europe remains the most prolific region, hosting the majority of the active countries listed (e.g., Germany, France, UK, Portugal, Italy, Belgium). Asia follows, driven primarily by China, with additional contributions from countries such as Indonesia and India. North America represents the third major cluster, led by the USA and Canada.

Table 1 shows the venues with more than 3 selected papers, including the famous conference in the field of MBRE, International Requirements Engineering Conference, and the well-known journal Requirements Engineering. Researchers can pay more attention to these venues to stay updated on the latest developments in the MBRE field. We have also compiled statistics on the citation counts of literature across different publication venues. These citation counts are sourced from Google Scholar, with statistics calculated as of January 2025. However, citation counts are highly influenced by temporal factors–since the publication years of the literature selected for each venue are not evenly distributed, deviations are inevitably present. Readers are therefore advised to use this information for reference only.

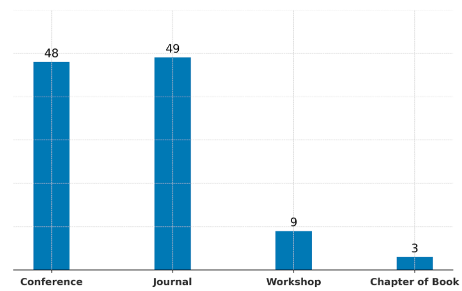
Fig. 4(a) shows that the majority of the selected papers are published by IEEE (36%), followed by Springer (32%), and Elsevier (28%). To further analyze the publication venues of the selected papers, we classified the selected research studies to assess their distribution by the type of publication (i.e. journal article, conference paper, workshop paper or chapter of book). As shown in Fig. 4(b), the most common

Table 1
Venues hosting more than three selected studies.

Venue	Number of Papers	Citations
IEEE International Requirements Engineering Conference (RE)	11	209
Requirements Engineering	7	515
Journal of Systems and Software	4	256
Information and Software Technology	4	223
Software and Systems Modeling	4	107
International Conference on Software Engineering (ICSE)	4	73
CIRP Design Conference	4	97
IEEE International Model-Driven Requirements Engineering Workshop (MoDRE)	3	88
Requirements Engineering: Foundation for Software Quality (REFSQ)	3	88



(a) Publisher



(b) Publication venue

Fig. 4. Publisher and Publication venue of selected paper.

publication types are journal articles (49 papers, 44.9%), and conference papers (48 papers, 44.0%) followed by workshop papers (9 papers, 8.3%). On the other hand, book chapters have only 3 occurrences (3 papers, 2.8%). Overall, journals and conferences are the most targeted publication venues, testifying that MBRE has become a significant research theme.

4.3. Research communities

In this section, we aim to answer the following research questions: RQ4. Which institutions and researchers have made outstanding contributions in the field of MBRE?

To identify the leading research institutions, we analyzed their contributions using the same dual-metric approach as the country-level analysis: Fractional Score (representing weighted contribution) and Number of Papers (representing total participation). Table 2 presents the top-performing institutions with a fractional score of 1.5 or higher.

The analysis reveals that Universidade Nova de Lisboa from Portugal is the most prominent institution in the MBRE field, leading with a significant fractional score of 5.6 from 7 participating papers. It is followed by the Technical University of Munich (Germany) with a score of 4.0 from 5 papers.

The data also highlights different institutional research profiles. For instance, Beihang University (China) participated in a notable four papers but achieved a fractional score of 1.9, indicating its frequent role as a collaborating partner in larger research teams. In contrast, institutions like the University of Zurich (Switzerland) and Nanjing University of Aeronautics and Astronautics (China) demonstrate a high contribution score relative to their publication volume (a score of 2.0 from just 2 papers each), suggesting more lead-authored or focused research efforts.

Other institutions making significant contributions include the University of Trento (Italy), Universitat Politècnica de València (Spain), and the University of Brighton (UK). Overall, the institutional landscape is predominantly led by European universities, complemented by key contributions from institutions in Asia, reflecting the global and collaborative nature of MBRE research.

Additionally, we identified 406 authors from the selected studies and constructed an author collaboration network using Gephi software. In

Table 2
Top active institutions in MBRE research ranked by fractional score (Score ≥ 1.5).

Institution	Score	Number of Papers
Universidade Nova de Lisboa	5.6	7
Technical University of Munich	4.0	5
University of Trento	2.6	4
Universitat Politècnica de València	2.4	4
University of Brighton	2.3	3
University of Zurich	2.0	2
Nanjing University of Aeronautics and Astronautics	2.0	2
Aalto University	2.0	2
Beihang University	1.9	4
Karlsruhe Institute of Technology	1.8	2
University of Manchester	1.7	3
Bandung Institute of Technology	1.5	2

Fig. 5, each node represents an author, with edges (connecting lines) indicating co-authorship relationships. Nodes with darker colors indicate higher degree in the adjacency graph, reflecting the author's extensive academic engagement within the field of model-based requirements engineering.

Based on this co-authored network diagram, we identified that there are several small research networks in this field, but no single large, unified network. Meanwhile, there are also several relatively prominent collaborative networks, and we can recognize the scholars who act as “pivots” within them. Through observation and analysis, scholars who play crucial roles in facilitating collaborations include (e.g., Y. Yang, J. Helming, M. Alférez, etc.) – selected based on prominent nodes in the graph) from various universities and research institutions. When further analyzing the co-authors of these “pivot” scholars, we found that research collaboration between universities and research institutions in the same geographical region occurs very frequently. Moreover, scholars who work in both universities and companies simultaneously, as well as those who have worked for different organizations in the past, tend to establish more extensive collaborative research relations.

In terms of the network's structural characteristics, this network consists of a considerable number of nodes and edges. The graph

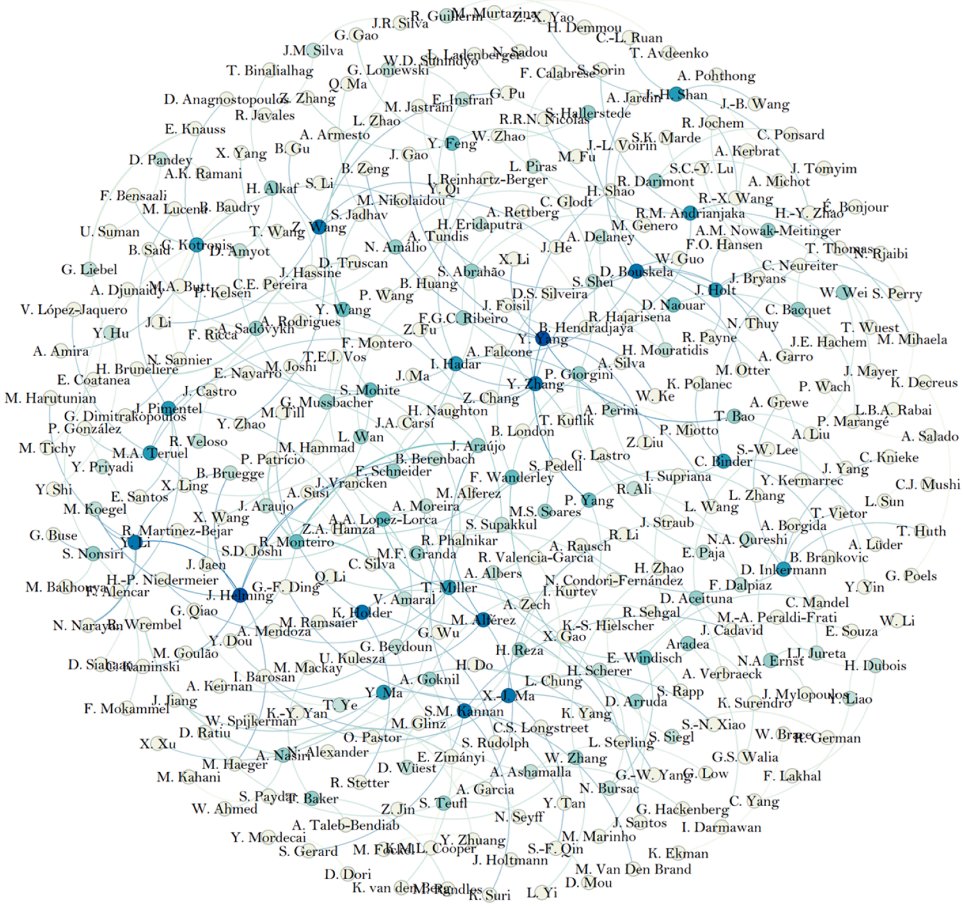


Fig. 5. Connections between authors (Darker nodes indicate a higher degree within the adjacency graph, representing authors who have published more academic works or engaged in more active scholarly exchanges and collaborations in the field of model-based requirements engineering).

modularity appears relatively high, indicating the network is divided into several distinct sub-networks with strong internal cohesion. However, the graph density is low, which implies that overall, the relationships between researchers tend to be fragmented, and research activities lack sufficient cross-subnetwork cooperation. For scholars engaged in this research field, it is necessary to pursue further mutual understanding and research collaboration in the future, as well as consistency in research methods. In this regard, we suggest that researchers should expand their knowledge accumulation and academic connections. Academic conferences, specialized workshops, or even online academic forums may serve as effective channels for achieving this goal.

4.4. Research topics and trends

In order to capture current trends and problems in MBRE research, we analyzed the topics of selected studies.

Table 3 presents the statistics based on the results of the analysis. First, model transformation and requirements analysis are the most active research directions, indicating the importance of these two types of techniques in the MBRE field.Second, the concentration of research in security requirements and agile requirements engineering reflects the focus on safety-critical systems and rapid development requirements. In addition, the topics of lightweight modeling, interdisciplinary requirements, and verticals such as automotive requirements engineering shown in the table indicate that the MBRE field is expanding into broader application areas but is still in its early stages. Finally, the low number of studies in areas such as toolchain support and requirements violation detection hints at some weaknesses in the practical application of MBRE. Researchers can obtain information about current domain focus, technology gaps, and potential future research directions.

To further identify the development trends of MBRE, we analyzed the changes in the number of literature pieces for each theme over time, with “approach evaluation” not participating in the theme trend discussion. As shown in Fig. 6, we found:

1). Requirements management and requirements change exhibit a downward trend, which may be due to the automation of MBRE-related tools (e.g., model version control, automated change impact analysis), reducing the need to research traditional manual requirements management processes. Meanwhile, the widespread adoption of agile development models in the industry emphasizes dynamic requirement iteration and rapid response, making traditional, strictly process-dependent requirements change management gradually replaced by lighter, model-driven continuous requirement evolution. The research focus may have shifted to how MBRE achieves real-time synchronization and incremental updates of requirement models in agile scenarios, rather than in-depth discussions on change management methodologies themselves.

2). Lightweight modeling and goal-oriented requirements engineering concentrated between 2010 and 2014, which may reflect the early exploratory integration of MBRE technologies with agile development (pursuing lightweight requirement representation for rapid iteration) and enterprise-level requirements governance. During this phase, researchers attempted to simplify complex requirement models into lightweight representations more acceptable to agile teams, but the implementation gap between the formal models relied on by goal-oriented requirements engineering and the informal requirement expressions (e.g., natural language descriptions of user stories) in actual agile practices likely led to a decline in subsequent research enthusiasm, reflecting

Table 3
Primary research topics of selected papers.

Topic	Number of Papers	Study ID
model transformation	11	S004,S023,S028,S033,S042,S043,S081,S082,S091,S095,S104
requirements analysis	10	S009,S012,S017,S048,S053,S062,S083,S101,S103,S105
requirements elicitation	8	S014,S035,S049,S054,S078,S084,S085,S089
requirements change	5	S034,S045,S056,S059,S079
requirements management	5	S019,S030,S046,S072,S076
security requirements	5	S044,S050,S055,S063,S087
approach evaluation	5	S013,S021,S037,S094,S108
agile requirements engineering	4	S040,S077,S096,S099
verification and validation	4	S020,S039,S069,S102
business-IT alignment	3	S015,S051,S070
goal-oriented requirements engineering	3	S010,S027,S041
interdisciplinary requirements	3	S005,S066,S107
lightweight modeling	3	S006,S016,S029
nonfunctional requirements	3	S001,S036,S065

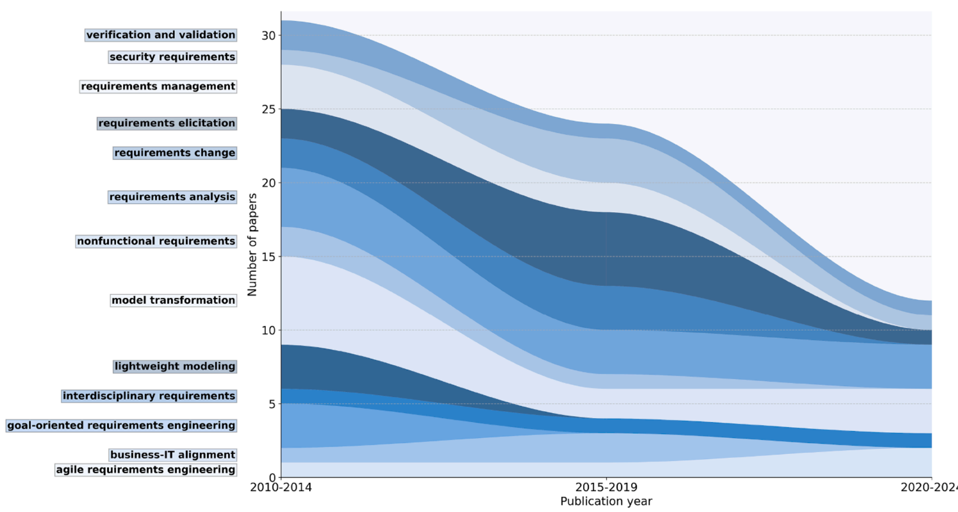


Fig. 6. Trends in research topics over times.

the phased exploration characteristics of early MBRE technology integration.

3). Requirements analysis and model transformation remain relatively stable and numerous, which reflect their role as the core foundation of the MBRE technical system. On the one hand, these technologies are the basis for the reliability of MBRE tools (ensuring semantic consistency in the analysis and transformation of requirement models) and need continuous optimization to adapt to new requirement representation paradigms. On the other hand, they possess cross-development-paradigm versatility (requiring model-based analysis and transformation of requirements in waterfall, agile, or AI-driven requirements engineering), making them long-standing “technical necessities” in MBRE research and supporting the continuous evolution of requirements engineering toward intelligent, model-driven approaches.

4). During the period of generally reduced literature quantity in 2020–2024, agile requirements engineering showed an upward trend. The divergent trend can be interpreted as the field’s strategic adaptation to external shocks and internal maturation. The COVID-19 pandemic acted as a catalyst, forcing distributed teams to seek more structured, model-based tools to replace collocated, informal agile practices, thereby accelerating the practical integration of MBRE. Concurrently, the research community’s focus shifted from questioning the feasibility of MBRE-Agile integration to solving its practical challenges, particularly in supporting large-scale agile development with techniques for hierarchical modeling and cross-team collaboration. This period reflects a consolidation phase: as MBRE shed its reputation for heaviness and embraced lighter, tool-supported practices, it found its niche not as a replacement for agile, but as a vital enabler for managing complexity, ensuring traceability, and maintaining consistency in ever-larger and more distributed agile projects.

Based on the above analysis, we can summarize the evolution logic of MBRE and answer RQ2. What are the key developmental phases and evolutionary characteristics in MBRE research?

- Early stage (2010–2014): MBRE explored integration with agile development and enterprise architecture to establish basic representation methods and transformation rules for requirement models.
- Mid-stage (2015–2019): Focused on the full lifecycle management of requirement models (incorporating changes and non-functional requirements) to support model-driven governance for complex projects.
- Late stage (2020–2024): MBRE deeply integrated with large-scale agile development, researching how to achieve cross-team collaboration in agile iterations through model-based requirements (e.g., model-driven DevOps requirement synchronization). Traditional themes (such as requirements management processes) declined due to tool automation and technological integration.

To further discuss the research topics, it is also meaningful to detect the keywords of the selected studies. Using the visualization tool TagCrowd, we created a tag cloud based on all the abstracts of 109 selected papers. Fig. 7 shows this tag cloud, from which we can get a general idea of the 50 most important keywords in the abstracts and their frequencies of occurrence. It is worth mentioning that we filtered out keywords such as “paper,” “study,” “present,” “propose,” “provide,” “describe,” “existing,” and other terms that frequently appear in abstracts but hold no substantial statistical value.

Fig. 7. Tag cloud of keywords of selected studies.

The tag cloud clearly reveals the research focus on model-based requirements engineering: high-frequency terms like “requirements” (615) and “modeling” (488) emphasize the centrality of requirement modeling, while “systems” (232), “development” (130), “approach” (151) and “method” (89) reflect that most MBRE research targets the exploration of MBRE methodologies within specific system contexts. Technical tools such as “model-based” (41), “formal” (54), and “tool” (61) reflect mainstream technologies, whereas “security” (39) and “change” (53) suggest under-explored areas like safety-critical systems and dynamic requirement management. Overall, research centers on complex software engineering needs, but non-functional requirements (e.g., “quality” 31) require further attention.

Fig. 8 shows the relationship between types and application domains using a two-dimensional heatmap. It highlights that general-purpose software systems are the most active area of MBRE research, especially in requirements modeling and analysis. Complex and collaborative systems also see significant research in these areas. Embedded and real-time systems focus more on requirements verification and evaluation, emphasizing the need for accuracy and reliability. While research in vertical domains like automotive, adaptive, and cloud computing systems is limited, these areas still show potential for requirements modeling and analysis. The blank spaces in the figure indicate research gaps, particularly in the lower-left corner, suggesting future research opportunities.

5. Technology systems

In this section, we aim to answer the following research questions: RQ5.What are the predominant technical approaches in MBRE, and what are their comparative strengths and weaknesses?

To address this, we first counted the modeling techniques used in the selected literature, and Table 4 demonstrates the modeling technologies with more than 2 occurrences. The result shows that UML (29 papers) and SysML (17 papers) are the most widely adopted technologies, highlighting the dominance of unified modeling languages in requirements engineering. The adoption of other techniques such as i* (8 papers), KAOS (6 papers), and GORE (5 papers) highlights the significance of goal-oriented modeling approaches in requirements engineering.

Next, we evaluated these modeling technologies through six dimensions to understand their effectiveness in practical applications, and these evaluation criteria are as follows.

- QC1[**Learnability**]: How intuitive is the technology for beginners to learn and use?
- QC2[**Expressiveness**]: Does the technology comprehensively describe functional/non-functional requirements and business process?
- QC3[**Collaboration**]: Does the technology support multi-user collaboration?
- QC4[**Toolchain**]: Are there mature tools and active communities supporting this technology?

Table 4
Mainly used modeling technology in selected studies.

Requirements Modeling Technology	Number of Papers
UML	29
SysML	17
i*	8
KAOS	6
DSL	6
GORE	5
OCL	3
BPMN	3
URN	2
NFR	2
Kano Model	2
Tropos	2
Techne	2

- QC5[**Scalability**]: Can the technology adapt to projects of different sizes?
- QC6[**Cost & Resources**]: What is the economic and time investment required to adopt this technology?

Table 5 describes our scoring criteria for these questions. To ensure the objectivity and reliability of the technical evaluation results, the scoring process was implemented as follows: the initial scoring of all modeling technologies against the six dimensions (QC1–QC6) was completed by the first author, who strictly referred to the technical details of each technology (extracted from the selected literature) and the specific scoring criteria defined in Table 5. After the initial scoring was finalized, the second author conducted a comprehensive review of the entire scoring dataset–specifically, cross-checking each score against the corresponding technology’s characteristics (e.g., toolchain maturity, collaboration support) and the predefined standards in Table 5 to confirm no inconsistencies or deviations. The final evaluation results were confirmed only after the second author verified that all scores aligned with the evaluation criteria.

Based on the results of the quality assessment (Fig. 9), the mainstream modeling technologies show significant scenario differentiation: general-purpose technologies (e.g., UML, BPMN) excel in requirements visualization and collaborative development by virtue of their mature tool chains (QC4 = 4) and high ease of learning (QC1 = 3-4); goal-driven approaches (e.g., KAOS, i*) are able to accurately decompose security goals (QC2 = 4), but are limited by steep learning curves (QC1 = 1-2) and lack of tool ecosystem (QC4 = 1-2). So the concentration of goal-oriented research in the early period (2010–2014) and its subsequent relative decline, as noted in Section 4.4, can be partly explained by our quality assessment. While these approaches offer strong expressiveness for goal decomposition, their steep learning curves and lack of a mature tool ecosystem likely hindered their widespread adoption and sustained research interest, especially as the community shifted focus towards more agile and practitioner-friendly methods. Domain specific language (DSL) shows high expressiveness in modeling specific business rules (QC2 = 4), but the need for custom development results in increased costs (QC6 = 2-3). Across all dimensions of the overall scores, it is evident that despite their strengths in capturing complex requirements, existing technologies present significant adoption hurdles due to fragmented tooling and inadequate support for teamwork. To overcome these barriers and broaden industrial adoption, the next generation of MBRE tools must prioritize cloud-based collaboration features and leverage AI to automate complex tasks such as model consistency checking and conflict resolution.

Based on the above analysis, we propose the following practical guidelines for practitioners seeking to adopt Model-Based Requirements Engineering (MBRE) in their projects. Our evaluation indicates that there is no “one-size-fits-all” modeling technology. Selection should be driven by specific project characteristics:

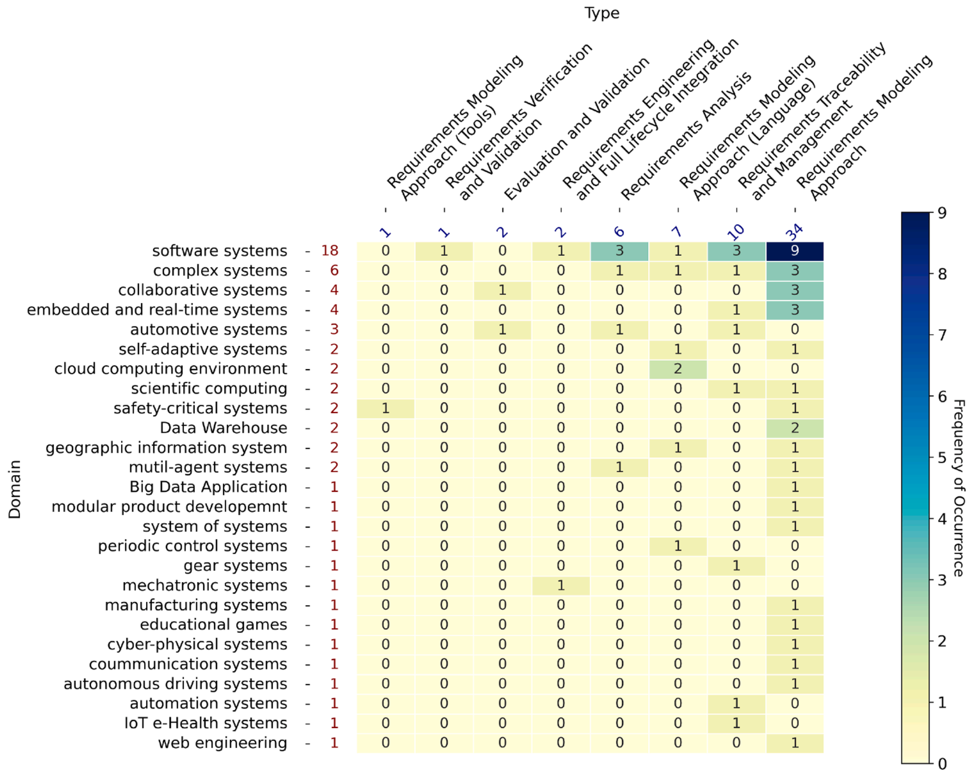


Fig. 8. Relationship between Type and Domain.

Table 5
Scoring criteria of modeling technology.

Dimension	4 (Excellent)	3 (Good)	2 (Fair)	1 (Poor)
Learnability	Intuitive; learn basics in 1 day	Core functions in 3–5 days	Needs 1–2 weeks training	Complex syntax; long learning
Expressiveness	Full coverage (functional/NFR/data)	Strong in one domain	Limited to single dimension	Text-dependent; poor visuals
Collaboration	Built-in version control & co-editing	File sharing & annotations	Requires 3rd-party tools	No collaboration features
Toolchain	Mature tools & active community	Dedicated niche tools	Basic open-source plugins	Manual drawing required
Scalability	Modular for all project sizes	Fit medium/large projects	For specific scales	High-cost scaling modifications
Cost & Resources	Free/open-source; minimal effort	Freemium model	Paid licenses	Custom dev/training needed

For General Software Systems and Agile Projects: Prioritize UML and lightweight modeling methods. They offer a gentle learning curve and mature toolchains, enabling effective support for rapid iteration and team collaboration.

For Safety-Critical and Embedded Real-Time Systems: SysML and formal methods are superior choices. They provide enhanced capabilities for expressing safety, reliability, and complex system architectures, despite a steeper learning curve.

For Goal-Driven and Strategic Alignment Projects: When requirements are tightly linked to high-level business objectives, goal-oriented approaches like i*/KAOS offer exceptional traceability and reasoning capabilities.

For Specific Vertical Domains (e.g., Healthcare, Manufacturing): Explore the use of Domain-Specific Languages (DSLs), as they can most accurately capture domain knowledge, though they require upfront development investment.

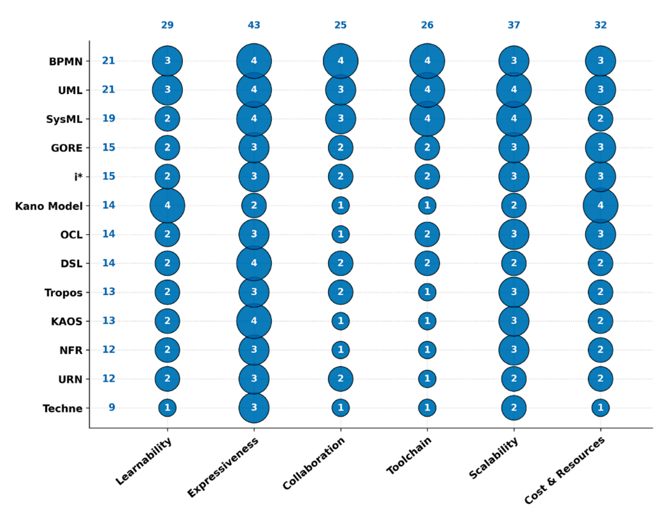


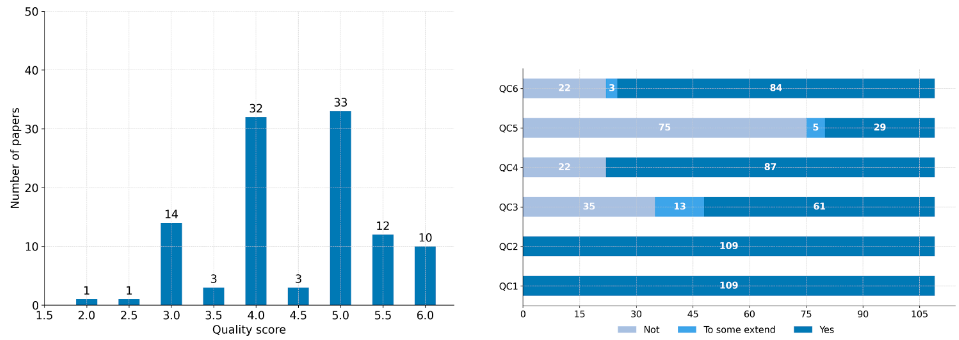
Fig. 9. Scoring result of RE modeling technologies.

6. Quality evaluation

In this section, we aim to answer the following research questions: RQ6.How are the qualities of the selected studies? To assess the quality of the selected literature, we adopted a six-dimensional quality assessment framework derived from Zhou et al. (2020) – a rigorously validated framework widely used in software engineering-related literature evaluations. We supplemented this framework with three clear scoring dimensions for consistency: Yes (fully meets the criterion) = 1, To some extent (partially meets the criterion) = 0.5, and No (does not meet the criterion) = 0. Detailed descriptions of each criterion and its corresponding scoring standards are provided in Table 6.

Table 6
Scoring criteria of each quality assessment criteria.

QC	Yes (1)	To some extent (0.5)	Not (0)
QC1	Clearly states context/framework	Mentions context vaguely without specifics	No context or framework mentioned
QC2	Objectives are specific and quantifiable	Objectives are general without technical details	No objectives stated
QC3	Step-by-step process with technical details	Brief procedure missing key steps	No procedure provided
QC4	Validated via experiments/case studies with quantitative data	Limited validation or subjective evaluation	No empirical validation
QC5	Clearly discusses limitations and improvement directions	Mentions limitations vaguely	No analysis of limitations
QC6	Specific future plans linked to current limitations	Vague directions without focus	No future directions discussed



(a) Frequency of quality score (b) Detailed scores of each criteria

Fig. 10. Scoring results of selected paper.

Additionally, it should be noted that the quality assessment criteria focus more on the quality of the paper's description (e.g., clarity, objectivity, structure) rather than the quality of the research itself. In other words, groundbreaking or high-level research does not necessarily receive high scores based on these standards.

Quality assessment criteria

- QC1: Is the industrial or methodological context of the MBRE research explicitly described?
- QC2: Are the objectives of the study clearly stated?
- QC3: Does the study provide a detailed procedure for applying the proposed MBRE method?
- QC4: Is the effectiveness of the proposed MBRE method validated empirically?
- QC5: Are the limitations of the proposed MBRE approach critically analyzed?
- QC6: Is there a discussion on the research direction for future work?

Fig. 10(a) shows the distribution of total quality scores across the 109 selected papers. It is evident that 33 studies received a score of 5, while 32 studies were rated with a score of 4. The average score for the papers was 4.49. The highest score of 6 was awarded to 10 paper, while the lowest score of 2 was given to just one paper. These results suggest that the majority of the studies are of good quality. Fig. 10(b) shows the detailed scores of each quality assessment criteria, it is clear that most of the selected papers performed well in presenting the research background, study objectives, evaluating the effectiveness of new findings, and outlining future directions. However, there remains room for improvement in the description of the MBRE steps and the discussion of limitations related to the proposed methods in several studies. Finally, we list the research studies with a quality score higher than 5 in Table 7.

7. Discussion

7.1. Comparison with prior reviews

To clarify the incremental contribution of this study, we compared our work with several recent systematic literature reviews (SLRs) in closely related areas of model-based requirements engineering (MBRE), as presented in Table 8. These prior SLRs focus on subfields of RE (e.g., requirements extraction in model-based systems engineering (MBSE), nonfunctional requirements (NFR) modeling, automated RE, and applications of large language models (LLMs) or visual analytics (VA) in RE). In summary, while prior SLRs advance specific subareas of RE, our work uniquely addresses the broader domain of MBRE–filling the gap of a systematic, cross-topic analysis of MBRE's research trends, technologies, and applications.

7.2. Threats to validity

To align with standard Systematic Literature Review (SLR) validity categories and address underdevelopment in this section, the following assesses threats across five core dimensions, linking directly to the study's methodology and data (Section 2–6) and evaluating their impact.

(1) Descriptive Validity

Descriptive validity refers to the accuracy of recording and describing objective processes (e.g., literature screening, data extraction) [4]. The primary threat here stems from single-reviewer bias: the entire literature screening process (from 437 initial papers to 192 eligible ones, then to 109 final papers) was completed independently by one author. While explicit inclusion/exclusion criteria (IC1–IC5, EC1–EC6) were used, subjective judgments about “MBRE relevance” for borderline studies (e.g., papers partially addressing requirements modeling but not explicitly labeling it as MBRE) may have led to accidental exclusion of relevant works or inclusion of marginally irrelevant ones. Additionally, data extraction (e.g., classifying application domains or modeling languages) lacked a strict operational definition for ambiguous cases (e.g., whether “AI-enabled software” belongs to “general software systems” or “vertical domains”), potentially causing minor misclassification in the final dataset. Impact: Medium–explicit criteria mitigated extreme bias, but single-reviewer operations introduced localized accuracy risks.

(2) Theoretical Validity

Theoretical validity concerns the rationality of the study's theoretical framework (research questions, evaluation dimensions). Two key threats exist: first, incomplete research question coverage: the 5W1H-based RQ1–RQ6 focus on traditional MBRE themes (motivations, publication trends, technical approaches) but omit emerging theoretical gaps (e.g., how MBRE integrates with generative AI for

Table 7
Research studies with a quality score higher than 5.

Paper Id	Year	Title	Score
S003	2010	A goal-based framework for contextual requirements modeling and analysis	6
S004	2010	Multi-view Composition Language for Software Product Line Requirements	6
S040	2014	Agile requirements engineering via paraconsistent reasoning	6
S045	2014	Change impact analysis for requirements: A metamodeling approach	6
S048	2014	CORAMOD: a checklist-oriented model-based requirements analysis approach	6
S055	2015	Modelling and reasoning about security requirements in socio-technical systems	6
S079	2019	An automated change impact analysis approach for User Requirements Notation models	6
S082	2019	Generating UML Use Case Models from Software Requirements Using Natural Language Processing	6
S097	2022	Formal requirements modeling for cyber-physical systems engineering: an integrated solution based on FORM-L and Modelica	6
S100	2024	RM4ML: requirements model for machine learning-enabled software systems	6
S014	2011	A domain specific requirements model for scientific computing: NIER track	5.5
S017	2011	User requirements modeling and analysis of software-intensive systems	5.5
S019	2011	Requirements Engineering for Scientific Computing: A Model-Based Approach	5.5
S021	2011	Evaluating requirements modeling methods based on user perceptions: A family of experiments	5.5
S031	2012	Towards model-driven game engineering for serious educational games: Tailored use cases for game requirements	5.5
S036	2013	Intention-oriented programming support for runtime adaptive autonomic cloud-based applications	5.5
S058	2015	Emotion-led modelling for people-oriented requirements engineering: The case study of emergency systems	5.5
S061	2016	An Extended TASM-Based Requirements Modeling Approach for Real-Time Embedded Software: An Industrial Case Study	5.5
S062	2016	Supporting agent oriented requirement analysis with ontologies	5.5
S064	2017	Model driven approach for real-time requirement analysis of multi-agent systems	5.5
S065	2017	Toward model-based requirement engineering tool support	5.5
S085	2019	Eliciting user requirements for e-collaboration systems: a proposal for a multi-perspective modeling approach	5.5

Table 8
Comparison with prior reviews.

Study	Topic	Time Range	Key Points of Interest	Research Method	Number of Papers
(Santos et al., 2025)	Requirements Extraction in MBSE	2000–2024	NL to Model, Arcadia/Capella method, Transformation between text and models	SLR	97
(Tariq and Cheema, 2021)	Nonfunctional Requirements Modeling	2006–2021	Elicitation, modeling of NFRs and their integration with FRs	SLR	25
(Umar and Lano, 2024)	Automated Requirements Engineering	1996–2022	automated RE tools, automation levels, tool evaluation methods	SLR	85
(Zadenoori et al., 2025)	Application of Large Language Models in Requirements Engineering	2023–2024	LLMs in RE activities, prompting strategies, evaluation methods	SLR	74
(Varikuti and Reddivari, 2024)	Application of Visual Analytics in Requirements Engineering	2010–2023	VA adoption in RE, VA tools/methods, Limitations of VA for RE	SLR	35
Ours	Model-Based Requirements Engineering	2010–2025	research hotspots, development trends, modeling technologies, application domains	SLR	109

requirement automation), limiting reflection of cutting-edge challenges. Second, narrow evaluation dimensions for modeling technologies: the six criteria (QC1–QC6: Learnability, Expressiveness, etc.) for assessing tools like UML/SysML do not include “cross-domain compatibility”–critical for interdisciplinary scenarios (e.g., healthcare-IoT integration)–underestimating practical limitations of these technologies in non-safety-critical domains. Impact: Medium–core frameworks address mainstream MBRE, but gaps reduce depth for future-oriented insights.

(3) Generalisability Validity

Generalisability refers to extending conclusions to other contexts. Threats include: database and language bias: literature was sourced from only four databases (IEEE Xplore, Springer Link, ScienceDirect, Google Scholar) and excluded non-English studies, missing region-specific research (e.g., Asian studies on smart manufacturing MBRE) and limiting applicability to non-English regions. Timeframe and grey literature exclusion: the 2010–2025 scope omits pre-2010 foundational MBRE work (e.g., early SysML v1.0 applications), and grey literature (industrial white papers, company technical reports) was excluded, omitting real-world industrial practices (,). domain skew: the 109 papers concentrate on general software systems (44.9%) and safety-critical systems (17.4%), with few on emerging domains (e.g., digital twins), reducing conclusions’ relevance to underrepresented fields. Impact: Medium-High–conclusions apply to mainstream academic contexts but not industrial or emerging domains.

(4) Interpretive Validity

Interpretive validity ensures conclusions logically derive from data. Threats include: trend attribution bias: the 2020–2024 publication decline was attributed primarily to COVID-19, but unaccounted factors (e.g., reduced software engineering funding, shifts to AI-driven research) may also contribute, leading to incomplete temporal trend understanding. Quality assessment limitations: the quality evaluation (QC1–QC6) focuses on “paper description clarity” (e.g., context, objectives) rather than “research rigor” (e.g., sample size for empirical studies), potentially overrating well-structured but methodologically weak papers and biasing interpretations of “high-quality MBRE research”. Impact: Low-Medium–quantitative data (e.g., 11 papers on model transformation) supports most trends, but subjective attribution introduces minor bias.

(5) Repeatability Validity

Repeatability validity ensures other researchers can replicate the study’s process and obtain consistent results. Key threats here stem from subjectivity in literature screening and classification (Section 2.2), with insufficient operational standardization. First, literature screening relied on a single author to narrow 192 pre-selected papers to 109 target studies. While inclusion/exclusion criteria (IC1–IC5, EC1–EC6) were defined, judging “MBRE relevance” for borderline papers (e.g., studies with implicit requirement modeling but no explicit “MBRE” labeling) depended on subjective interpretation–no multi-reviewer cross-validation or decision logs for ambiguous cases exist, leading to potential variance in replicated sample sizes. Second, classifying the 192 papers into Type A (theoretical) to Type D (empirical) and research topics (e.g., “model transformation”) lacked strict boundaries. For example, studies overlapping theoretical and methodological content were categorized by subjective “core contribution” judgment, and cross-thematic papers were assigned to one topic only. This ambiguity may cause inconsistent classification in replication, altering trend statistics (e.g., counts of key research topics). Impact: Medium-High. Subjective screening and unclear classification rules reduce the reproducibility of the 109-paper dataset and trend analyses, weakening result reliability.

7.3. Conclusion

This survey comprehensively analyzes the developments and trends in Model-Based Requirements Engineering (MBRE) from 2010 to 2025 based on 109 selected studies, revealing key research hotspots and evolutionary characteristics. Model transformation (11 papers) and requirements analysis (10 papers) are identified as the most active research directions, forming the core technical foundation of MBRE, with their stability over time reflecting the need for continuous optimization to adapt to new requirement representation paradigms and cross-development-paradigm versatility. Security requirements and agile requirements engineering have gained increasing attention, reflecting the focus on safety-critical systems and rapid development needs, while lightweight modeling and goal-oriented requirements engineering were concentrated in the early stage (2010–2014), showing the exploratory integration of MBRE with agile development and enterprise architecture at that time.

In terms of development phases, MBRE explored the integration with agile development and established basic modeling methods in the early stage (2010–2014); shifted to full lifecycle management of requirement models (including changes and non-functional requirements) in the mid-stage (2015–2019); and deeply integrated with large-scale agile development in the late stage (2020–2024), with agile requirements engineering showing an upward trend despite the overall decline in publications during 2020–2024, indicating the acceleration of MBRE’s adaptation to agile paradigms.

The application domains of MBRE are mainly concentrated in general software systems, with significant research in requirements modeling and analysis, while embedded and real-time systems focus more on requirements verification and evaluation. Emerging domains such as automotive, cloud computing, and AI-enabled systems show potential but remain underdeveloped, with visible research gaps in the two-dimensional heatmap of Type-Application Domain.

Regarding the global research landscape, the updated demographic analysis highlights the highly collaborative nature of the MBRE community. Europe remains the most prolific region, hosting prominent institutions such as Universidade Nova de Lisboa and Technical University of Munich. At the national level, China and Germany emerge as the dominant leaders; China leads in weighted contribution (fractional score), indicating a strong output of lead-authored research, while Germany holds the highest participation volume, reflecting extensive collaborative activity.

Despite notable progress in both research and practical applications, there are still several challenges. First, the quality disparity among the reviewed studies, including inconsistent methodological rigor and the lack of empirical validation in some studies, indicates a need for more robust and reproducible research in this field. Additionally, the analysis reveals gaps in interdisciplinary research and issues related to the integration of MBRE with emerging technologies, such as AI-based tools and low-code platforms.

The future of MBRE lies in overcoming these limitations by enhancing the integration of toolchains (e.g., through low-code platforms and AI-driven automation), deepening interdisciplinary research, and expanding applications in emerging technologies and vertical domains. Strengthening collaboration between academia and industry to develop more robust, scalable, and automated MBRE solutions will be crucial for its sustained advancement.

CRediT authorship contribution statement

Zhongpei Zhao: Writing – original draft, Formal analysis; **Zhengshu Zhou:** Writing – review & editing, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by Tianjin Natural Science Foundation Project (No.25JCQNJC00060).

Appendix A. Selected studies

See Table A.9.

Table A.9
Selected studies in the survey.

Paper Id	Publication Year	Title
S001	2010	Engineering dependability requirements for complex systems – A new information model definition
S002	2010	A Model for Requirements Traceability in a Heterogeneous Model-Based Design Process: Application to Automotive Embedded Systems
S003	2010	A goal-based framework for contextual requirements modeling and analysis
S004	2010	Multi-view Composition Language for Software Product Line Requirements
S005	2010	Towards a unified Requirements Modeling Language
S006	2010	Very Lightweight Requirements Modeling
S007	2010	An Effective Requirement Engineering Process Model for Software Development and Requirements Management
S008	2010	Using VCL as an Aspect-Oriented Approach to Requirements Modelling
S009	2010	Case-based analysis in user requirements modelling for knowledge construction
S010	2010	Mdgore: Towards Model-Driven and Goal-Oriented Requirements Engineering
S011	2010	SPARDL: A Requirement Modeling Language for Periodic Control System
S012	2010	Model Based Requirements Analysis and Testing of Automotive Systems with Timed Usage Models
S013	2010	Requirements Modeling with the Aspect-oriented User Requirements Notation (AoURN): A Case Study
S014	2011	A domain specific requirements model for scientific computing: NIER track
S015	2011	A Goal-Oriented Requirements Engineering Method for Business Processes
S016	2011	Flexible Sketch-Based Requirements Modeling
S017	2011	User requirements modeling and analysis of software-intensive systems
S018	2011	An architecture-oriented model-driven requirements engineering approach
S019	2011	Requirements Engineering for Scientific Computing: A Model-Based Approach
S020	2011	Evaluating the use of model-based requirements verification method: A feasibility study
S021	2011	Evaluating requirements modeling methods based on user perceptions: A family of experiments
S022	2012	Towards a Requirements Modeling Language for Self-Adaptive Systems
S023	2012	Deriving software architectural models from requirements models for adaptive systems: the STREAM-A approach
S024	2012	A modeling language to support early lifecycle requirements modeling for systems engineering
S025	2012	BRM–A Behavior Based Requirements Modeling Method
S026	2012	The RE-Tools: A multi-notational requirements modeling toolkit
S027	2012	Model-Driven Development for Requirements Engineering: The Case of Goal-Oriented Approaches
S028	2012	Using machine learning to enhance automated requirements model transformation
S029	2012	Flexible, lightweight requirements modeling with Flexisketch
S030	2012	Requirements management within a full model-based engineering approach
S031	2012	Towards model-driven game engineering for serious educational games: Tailored use cases for game requirements
S032	2012	Analyzing the understandability of Requirements Engineering languages for CSCW systems: A family of experiments
S033	2013	PLANT: A pattern language for transforming scenarios into requirements models
S034	2013	Model-based approach for change propagation analysis in requirements
S035	2013	Feature-oriented stigmergy-based collaborative requirements modeling: an exploratory approach for requirements elicitation and evolution based on web-enabled collective intelligence

Table A.9
Continued.

Paper Id	Publication Year	Title
S036	2013	Intention-oriented programming support for runtime adaptive autonomic cloud-based applications
S037	2013	MIRA: A tooling-framework to experiment with model-based requirements engineering
S038	2013	Comparing the comprehensibility of requirements models expressed in Use Case and Tropos: Results from a family of experiments
S039	2014	SnapMind: A framework to support consistency and validation of model-based requirements in agile development
S040	2014	Agile requirements engineering via paraconsistent reasoning
S041	2014	Modeling the requirements for big data application using goal oriented approach
S042	2014	A requirements engineering methodology combining models and controlled natural language
S043	2014	Model-based requirement generation
S044	2014	INCREMENT: A Mixed MDE-IR Approach for Regulatory Requirements Modeling and Analysis
S045	2014	Change impact analysis for requirements: A metamodeling approach
S046	2014	A method and tool for tracing requirements into specifications
S047	2014	Requirement and interaction analysis using aspect-oriented modeling
S048	2014	CORAMOD: a checklist-oriented model-based requirements analysis approach
S049	2015	UML-based requirement modeling of Web online synchronous collaborative public participatory GIS
S050	2015	Developing a Novel Holistic Taxonomy of Security Requirements
S051	2015	Model-Based Requirements Engineering for Data Warehouses: From Multidimensional Modelling to KPI Monitoring
S052	2015	A semantic web enabled approach to reuse functional requirements models in web engineering
S053	2015	Extension Innovation Design of Product Family Based on Kano Requirement Model
S054	2015	Product Requirement Modeling and Optimization Method Based on Product Configuration Design
S055	2015	Modelling and reasoning about security requirements in socio-technical systems
S056	2015	Efficient Impact Analysis of Changes in the Requirements of Manufacturing Automation Systems
S057	2015	A Model-Based Approach for Requirements Engineering for Systems of Systems
S058	2015	Emotion-led modelling for people-oriented requirements engineering: The case study of emergency systems
S059	2016	Requirements change management based on object-oriented software engineering with unified modeling language
S060	2016	A Modular Requirements Engineering Framework for Web-Based Toolchain Integration
S061	2016	An Extended TASM-Based Requirements Modeling Approach for Real-Time Embedded Software: An Industrial Case Study
S062	2016	Supporting agent oriented requirement analysis with ontologies
S063	2017	A Security Requirements Modelling Language to Secure Cloud Computing Environments
S064	2017	Model driven approach for real-time requirement analysis of multi-agent systems
S065	2017	Toward model-based requirement engineering tool support
S066	2017	Transforming Multidisciplinary Customer Requirements to Product Design Specifications
S067	2017	Integration of Self-adaptation Approach on Requirements Modeling
S068	2017	Model Based Requirements Engineering for the Development of Modular Kits
S069	2017	CoSTest: A Tool for Validation of Requirements at Model Level

Table A.9
Continued.

Paper Id	Publication Year	Title
S070	2017	Model-based requirements engineering: Architecting for system requirements with stakeholders in mind
S071	2017	Requirements Engineering for Data Warehouses (RE4DW): From Strategic Goals to Multidimensional Model
S072	2017	Model-Based Requirements Management in Gear Systems Design Based On Graph-Based Design Languages
S073	2017	Towards Industry 4.0: Gap Analysis between Current Automotive MES and Industry Standards Using Model-Based Requirement Engineering
S074	2018	Model-based requirements specification of real-time systems with UML, SysML and MARTE
S075	2018	A Model-based Approach for Managing Criticality Requirements in e-Health IoT Systems
S076	2019	Model-Based Requirement Engineering to Support Development of Complex Systems
S077	2019	An Ontology-Based Approach to the Agile Requirements Engineering
S078	2019	A Chatbot for Goal-Oriented Requirements Modeling
S079	2019	An automated change impact analysis approach for User Requirements Notation models
S080	2019	A new Requirements Engineering approach for Manufacturing based on Petri Nets
S081	2019	RM2PT: A Tool for Automated Prototype Generation from Requirements Model
S082	2019	Generating UML Use Case Models from Software Requirements Using Natural Language Processing
S083	2019	Requirements Dependency Graph Modeling on Software Requirements Specification Using Text Analysis
S084	2019	Constructing True Model-Based Requirements in SysML
S085	2019	Eliciting user requirements for e-collaboration systems: a proposal for a multi-perspective modeling approach
S086	2019	Use, potential, and showstoppers of models in automotive requirements engineering
S087	2020	A security requirements modelling language for cloud computing environments
S088	2020	Applying Acceptance Requirements to Requirements Modeling Tools via Gamification: A Case Study on Privacy and Security
S089	2021	A WebGIS Interface Requirements Modeling Language
S090	2021	Modeling functional requirements using tacit knowledge: a design science research methodology informed approach
S091	2021	Automatic generation of Web service for the Praxeme software aspect from the ReLEL requirements model
S092	2021	Enabling model-based requirements engineering in a complex industrial System of Systems environment
S093	2021	Towards the Integration of Cybersecurity Risk Assessment into Model-based Requirements Engineering
S094	2021	Applying Model-based Requirements Engineering in Three Large European Collaborative Projects: An Experience Report
S095	2022	RM2Doc: A Tool for Automatic Generation of Requirements Documents from Requirements Models
S096	2022	Approach for model-based requirements engineering for the planning of engineering generations in the agile development of mechatronic systems
S097	2022	Formal requirements modeling for cyber-physical systems engineering: an integrated solution based on FORM-L and Modelica
S098	2023	MBIPV: a model-based approach for identifying privacy violations from software requirements
S099	2023	Aspects of modelling requirements in very-large agile systems engineering
S100	2024	RM4ML: requirements model for machine learning-enabled software systems
S101	2024	Multicriteria requirement ranking based on uncertain knowledge representation and reasoning
S102	2024	Requirements Structure for System Requirements Formal Modelling, Verification and Validation

Table A.9
Continued.

Paper Id	Publication Year	Title
S103	2024	Identifying Uncertainty in Large-Scale Industry 4.0 Software Projects Through Model-Based Requirements Engineering
S104	2024	Requirements Modeling and Automatic Transformations Towards Autonomous Driving Scenarios Description
S105	2024	User requirement modeling and evolutionary analysis based on review data: Supporting the design upgrade of product attributes
S106	2024	Towards model-based requirements engineering: Construction method of stakeholder value networks model based on MBSE
S107	2024	A Model-Driven Requirements Engineering Method for Human-Centered Digitalisation of Agriculture
S108	2024	An iterative approach for model-based requirements engineering in large collaborative projects: A detailed experience report
S109	2025	OCLVerifer: Automated verification of OCL contracts in requirements models

References

Abrahão S., Insfran E., Carsí J.A., Genero M., Evaluating requirements modeling methods based on user perceptions: A family of experiments, Inf. Sci. (N.Y.) 181 (16) (2011) 3356–3378.

Aceituna D., Do H., Walia G.S., Lee S.-W., Evaluating the use of model-based requirements verification method: A feasibility study, In: Workshop on Empirical Requirements Engineering (EmpiRE 2011), 2011, pp. 13–20.

Alférez M., Santos J., Moreira A., Garcia A., Kulesza U., Araújo J., Amaral V., Multi-view Composition Language for Software Product Line Requirements, In: Software Language Engineering, 2010, pp. 103–122.

Ali R., Dalpiaz F., Giorgini P., A goal-based framework for contextual requirements modeling and analysis, Require. Eng. 15 (4) (2010) 439–458.

Ameller D., Franch X., Gómez C., Martínez-Fernández S., Araújo J., Biffl S., Cabot J., Cortellessa V., Fernández Daniel Méndez, Moreira A., Muccini H., Vallecillo A., Wimmer M., Amaral V., Böhm W., Bruneliere H., Burgueño L., Goulão M., Teufl S., Berardinelli L., Dealing with non-functional requirements in model-Driven development: a survey, IEEE Trans. Softw. Eng. 47 (4) (2021) 818–835.

Bacquet C., Marangé P., Bonjour É., Kerbrat A., Requirements structure for system requirements formal modelling, verification and validation, In: 18th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2024, 2024, pp. 289–294.

Bao T., Yang J., Yang Y., Yin Y., RM2Doc: A Tool for Automatic Generation of Requirements Documents from Requirements Models, In: 2022 IEEE/ACM 44th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion), 2022, pp. 188–192.

Brace W., Ekman K., CORAMOD: A checklist-oriented model-based requirements analysis approach, Require. Eng. 19 (1) (2014) 1–26.

Goknil A., Kurtev I., van den Berg K., Spijkerman W., Change impact analysis for requirements: A metamodeling approach, Inf. Softw. Technol. 56 (8) (2014) 950–972.

Granda M.F., Condori-Fernández N., Vos T.E.J., Pastor O., CoSTest: A Tool for Validation of Requirements at Model Level, In: 2017 IEEE 25th International Requirements Engineering Conference (RE), 2017, pp. 464–467.

Hadar I., Reinhartz-Berger I., Kuflik T., Perini A., Ricca F., Susi A., Comparing the comprehensibility of requirements models expressed in Use Case and Tropos: Results from a family of experiments, Inf. Softw. Technol. 55 (10) (2013) 1823–1843.

Holder K., Zech A., Ramsaier M., Stetter R., Niedermeier H.-P., Rudolph S., Till M., Model-Based Requirements Management in Gear Systems Design Based On Graph-Based Design Languages, Appl. Sci. 7 (11) (2017) 1112.

Li Y., Peng R., Wang B., Challenges in context-aware requirements modeling: a systematic literature review, In: Requirements Engineering for Internet of Things, 2018, pp. 140–155.

Liao Y., Yang C., Li W., Extension Innovation Design of Product Family Based on Kano Requirement Model, In: 3rd International Conference on Information Technology and Quantitative Management, ITQM 2015, 2015, pp. 268–277.

Ma Y., Dou Y., Xu X., Jiang J., Yang K., Tan Y., Multicriteria requirement ranking based on uncertain knowledge representation and reasoning, Adv. Eng. Inf. 59 (2024) 102329.

Kannan S.M., Suri K., Cadavid J., Barosan I., Van Den Brand M., Alferez M., Gerard S., Towards Industry 4.0: Gap Analysis between Current Automotive MES and Industry Standards Using Model-Based Requirement Engineering, In: 2017 IEEE International Conference on Software Architecture Workshops (ICSAW), 2017, pp. 29–35.

Mouratidis H., Shei S., Delaney A., A security requirements modelling language for cloud computing environments, Softw. Syst. Model. 19 (2) (2020) 271–295.

Pandey D., Suman U., Ramani A.K., An Effective Requirement Engineering Process Model for Software Development and Requirements Management, In: 2010 International Conference on Advances in Recent Technologies in Communication and Computing, 2010, pp. 287–291.

Rashid M., Anwar M.W., Azam F., Kashif M., Model-based requirements and properties specifications trends for early design verification of embedded systems, In: 2016 11th System of Systems Engineering Conference (SoSE), 2016, pp. 1–7.

Santos J.L., Martins L.E.G., Molléri J.S., Requirements extraction from model-based systems engineering: A systematic literature review, J. Syst. Softw. 226 (2025) 112407.

Supakkul S., Chung L., The RE-Tools: A multi-notational requirements modeling toolkit, In: 2012 20th IEEE International Requirements Engineering Conference (RE), 2012, pp. 333–334.

Tariq S., Cheema S.M., Approaches for Non-Functional Requirement Modeling: A Literature Survey, In: 2021 4th International Conference on Computing & Information Sciences (ICCIS), 2021, pp. 1–6.

Umar M.A., Lano K., Advances in automated support for requirements engineering: A systematic literature review, Require. Eng. 29 (2) (2024) 177–207.

Varikuti S., Reddivari S., Visual Analytics in Requirements Engineering: A Systematic Literature Review, In: 2024 IEEE International Conference on Information Reuse and Integration for Data Science (IRI), 2024, pp. 320–325.

Wanderley F., Silva A., Araujo J., Silveira D.S., SnapMind: A framework to support consistency and validation of model-based requirements in agile development, In: 2014 IEEE 4th International Model-Driven Requirements Engineering Workshop (MoDRE), 2014, pp. 47–56.

Wang Z., Li J., Zhao Y., Qi Y., Pu G., He J., Gu B., SPARDL: A Requirement Modeling Language for Periodic Control System, In: Leveraging Applications of Formal Methods, Verification, and Validation, 2010, pp. 594–608, Berlin, Heidelberg. Springer.

Whittle J., Hutchinson J., Rouncefield M., Burden H., Heldal R., A taxonomy of tool-related issues affecting the adoption of model-driven engineering, Softw. Syst. Model. 16 (2) (2017) 313–331.

Windisch E., Mandel C., Rapp S., Bursac N., Albers A., Approach for model-based requirements engineering for the planning of engineering generations in the agile development of mechatronic systems, In: 32nd CIRP Design Conference (CIRP Design 2022) - Design in a Changing World, 2022, pp. 550–555.

Yang Y., Li X., Liu Z., Ke W., RM2PT: A Tool for Automated Prototype Generation from Requirements Model, In: 2019 IEEE/ACM 41st International Conference on Software Engineering: Companion Proceedings (ICSE-Companion), 2019, pp. 59–62.

Yang Y., Zeng B., Gao J., RM4ML: Requirements model for machine learning-enabled software systems, Require. Eng. (2024).

Yang Z., Li Z., Jin Z., Chen Y., A Systematic Literature Review of Requirements Modeling and Analysis for Self-adaptive Systems, In: Requirements Engineering: Foundation for Software Quality, 2014, pp. 55–71.

Zadenoori Mohammad Amin, Dąbrowski J., Alhoshan W., Zhao L., Ferrari A., Large Language Models (LLMs) for Requirements Engineering (RE): A Systematic Literature Review, 2025.

Zhou Z., Zhi Q., Morisaki S., Yamamoto S., A Systematic Literature Review on Enterprise Architecture Visualization Methodologies, IEEE Access 8 (2020) 96404–96427.